Method Statement

Ref. #: DCP00/08-0198-A-2025



DonProof L

(High-performance, pre-applied, loose laid PVC membrane for below-grade waterproofing)

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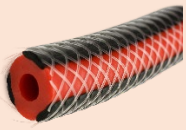
Section A: General Comments

General Notes:

The information below is a detailed overview of the application of DCP's **DonProof L** waterproofing membrane and should be read in conjunction with the relevant technical data sheet prior to application. All DCP Products should be applied by experienced specialist contractors.

All the points below assume the correct preparation of the relevant surface.

System Products:

Separation/protection layers below Membrane: **Geotex** (weighing no less than 350 g/m²)		Waterproofing membrane: **DonProof L**	
Flanges/sockets for resin Injection: **DonProof PSocket PVC**		PVC disk: **DonProof PDisc**	
PVC Waterstop: **Hydrostop range**		Swellable water bar: **Hyperseal DP-2010H**	
Re-injectable hose: **Flexseal HI300**		Polypropelyne protection boards: **Plasti-board**	

Tools and Equipment:

It is suggested that the following list of equipment are adopted as a minimum requirement

Personal protection	:	*Protective overalls*
	:	*Goggles or a face mask*
	:	*Good quality gloves*
	:	*Safety shoes*
	:	*Safety helmet*
Application equipment	:	*Manual hot-air blower (Fig.1)*
	:	*Automatic hot wedge welder (Fig.2)*
	:	*Utility knife with blade (Fig.3)*
	:	*Pressure roller (Fig.4)*
	:	*Nail gun + Nails (Fig.5)*
	:	*Measuring & marking tools (Fig.6)*





Fig.1: Manual hot-air blower



Fig.2: Automatic hot wedge welder



Fig.3: Utility knife with blade



Fig. 4: Pressure roller



Fig. 5: Nail gun + Nails



Fig.6: Measuring & marking tools

Section B: Application

1.0 Substrate Preparation

1.1 After completing the excavation, pile head removal, and trimming of the existing ground to the final formation level, the Engineer will inspect and approve the formation level before moving forward. Once approved, the first blinding and waterproofing layer can begin.

1.2 A concrete blinding mix will be poured, keeping a 100mm gap from the pile face. This concrete will be finished with a steel trowel, and all edges and corners will be rounded with a minimum radius of 50mm.

1.3 Any surface protrusions in the concrete base will be removed by chiseling or grinding. Nails, wires, loose stones, and similar debris will be cleared, and all voids, service openings, and holes will be properly filled to match the surface level.



1.4 Any accumulated water will be extracted using wet/dry vacuums and air compressors.

1.5 Ensure all substrates are solid, clean, and free of debris, sharp edges, or voids.

1.6 Detailing of all critical points i.e. Up-stands, penetrations, ground anchors, pile caps, etc., must be completed before installing the primary waterproofing membrane.

2.0 Separation/Protection Layers below the Membrane

2.1 **Horizontal application**

2.1.1 A cushion layer of geotextile membrane, weighing no less than 350 g/m², shall be installed directly over the blinding concrete, and walls shotcrete as part of the preparation process.

2.1.2 Prior to placement, surfaces should be thoroughly cleaned to ensure they are free from any loose materials, sharp projections, or debris that could damage the membrane.

2.1.3 For horizontal application, the geotextile will be laid loosely across the blinding concrete and positioned to provide a minimum overlap of 100 mm between adjacent sheets, ensuring full coverage of the underlying blinding concrete. In order to avoid displacement of geotextile sheets, fasten the overlaps with spot welding using a manual welding heat gun.



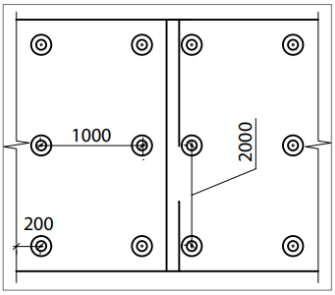


2.2 For vertical application, the geotextile can be fixed using the fixing discs with a minimum overlap of 100mm between adjacent sheets. In order to avoid displacement of geotextile sheets, fasten the overlaps with spot welding using a manual welding heat gun.

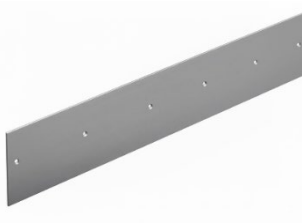
2.3 Fixing the discs is by nail gun (e.g. Hilti DX 36) into shotcrete substrates, or by fixing spikes into pre-drilled holes in reinforced concrete substrates. **Donproof L** sheet membranes get fixed by heat welding them on to the surface of these fixing discs.







2.4 Alternatively, the geotextile can be fixed using spikes driven into pre-drilled holes with a minimum overlap of 100mm between adjacent sheets. The main **Donproof L** membrane is then heat welded directly onto the surface of these fixing plates.



3.0 Waterproofing Membrane - Horizontal Application

3.1 After laying the geotextile membrane, it is essential to inspect the surface for any loose debris or sharp projections before proceeding with the installation of the PVC waterproofing membrane.

3.2 Consideration should be given to irregular shapes of basement slabs before determining the direction and sequence of membrane laying.

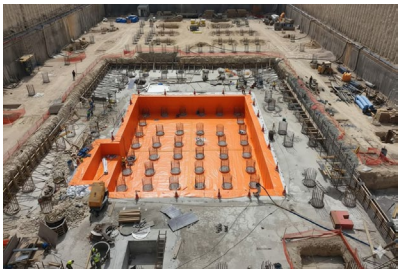
3.3 **DonProof L** should be cut to a convenient length required for each working zone ensuring there is a minimum overlap of 80mm for membrane continuity.

3.4 Membrane is unrolled always with the black colour side facing the prepared substrate/geotextile and the other face towards the concrete to be poured.

3.5 Signal layer shows damage and aids visual inspection.

3.6 Unroll and position the PVC membrane, ensuring sufficient overlap at the ends and edges to intersect with the vertical waterproofing.





3.7 Depending on the location, weld the PVC membrane onto the pile head PVC waterstop by at least 30-40mm, leaving 10 - 15mm of the waterstop exposed from the top edge. Special Leister welding machines should be used for welding joints.





3.8 All seam overlaps of the PVC membranes should be a minimum of 80mm, with the width of the finished welded seam being > 30mm for single seams.

3.9 The underside of the succeeding sheet and upper side of the PVC membrane already laid should be clean, dry and free from any contamination before welding using automatic welding machine.

3.10 Prior to any heat welding work, it is important to conduct a welding test on a similar PVC membrane specimen.



3.11 Using a hand roller, ensure a watertight seal by over rolling the overlaps, end laps and cut edges.

3.12 Inspect and test all joints of the PVC membrane to ensure their soundness and adherence.

4.0 Compartmentalization

4.1 **Waterstops:**

4.1.1 Shop drawings illustrating the construction joint locations and positioning of the compartmentalization waterstops should be prepared by the applicator and submitted for approval by the consultant.

4.1.2 These waterstops should be profiled with ribs on one side (rear fix).

4.1.3 After the completion of **DonProof L** laying and testing, **Hydrostop PVC** waterstops are used for compartmentalization.

4.1.4 lay down **Hydrostop PVC** waterstops along the line of construction joints and compartments with the flat side facing the membrane surface and the ribs facing the concrete.

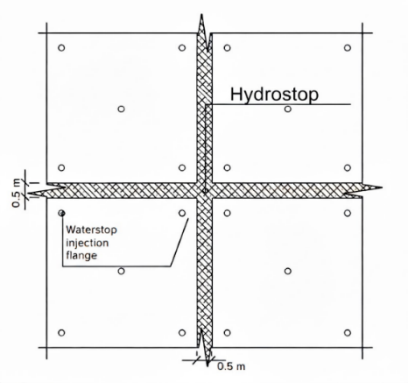
4.1.5 Weld the edges of the waterstop to the membrane with manual or semiautomatic equipment to form closed compartments.

4.1.6 Compartment size should be 120- 150 m^2 depending on the site condition.

4.1.7 All welds will be continuously inspected to ensure no voids, no pinholes, etc. are allowed.

4.2 **Injection system:**

4.2.1 After the waterproofing is compartmentalized, install an injection system consisting of injection flanges and hoses that are designed to repair waterproofing in case any leaks appear.



4.2.2 It is recommended to install 5 injection flanges into each compartment.

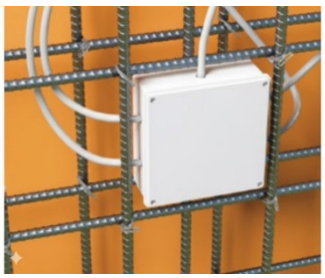
4.2.3 Normally, one out of five flanges installed in the center of a section, and the remaining ones — in corners at a maximum distance of 1.5 m from the edges.

4.2.4 **Donproof PSocket** injection flanges shall be welded to the membrane surface with the hot air hand tool. Connect an injection hose to the flange through connecting fitting.

4.2.5 The hoses should be terminated in a junction box.







5.0 Protection Layer

5.1 **Geotextile:**

5.1.1 Following the installation of the PVC waterproofing membrane and waterstops and flages, a second layer of geotextile will be laid over the PVC membrane as a protection layer with 100 mm overlaps, , this will prevent mechanical damage to the membrane.

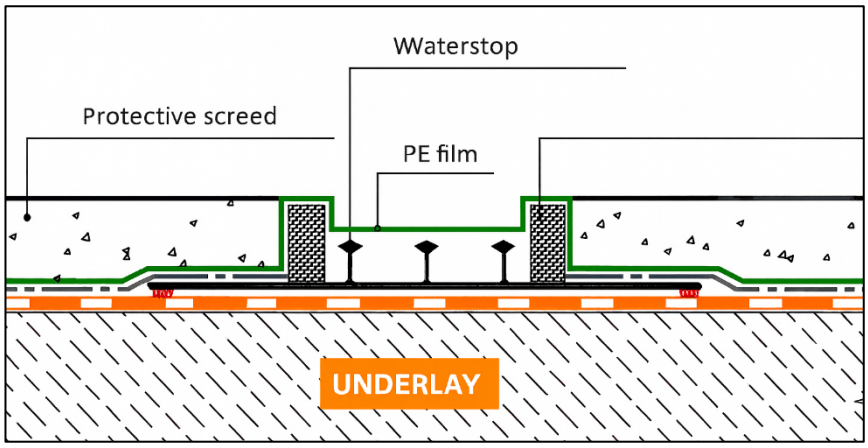
5.1.2 Lay geotextile within one section/compartment limited by waterstops, waterstops must remain exposed their fins must be open for subsequent concrete casting.

5.2 **Protective screed:**

5.2.1 A protective screed of at least 50 mm should be installed to prevent the membrane from mechanical damage during reinforcement installation.

5.2.2 Before the installation of the protective screed, place the formwork that will limit contamination of the waterstop with the protective screed.

5.2.3 Keep protective screed away from the waterstop fins. To do this, first cover all waterproofing, including waterstops, with polyethylene film, and after the protective screed is placed, cut off PE film above the waterstop.



6.0 Waterproofing Membrane - Vertical Application

6.1 Prior to casting the walls, the control and injection flanges shall be fixed on the formwork, connected with the hoses and terminated at the junction box.

6.2 **Flexseal HI300** re-injectable hose shall be installed at all construction joints of the walls and terminated at the junction box.

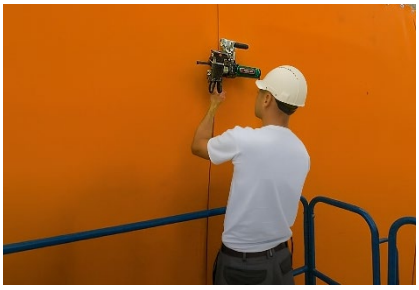


6.3 Fix **Hyperseal DP-2010H** swellable bars at the construction joints.

6.4 After casting of the wall, the geotextile shall be fixed on the concrete wall with the discs as per section 2.

6.5 After completing the horizontal waterproofing, a 50 x 50 mm sand-cement fillet will be applied at the interface between the horizontal and vertical surfaces, on top of the screed.

6.6 **DonProof L** membrane shall be fixed on the vertical surface and welded to the fixing discs, overlap shall be of minimum 80 mm double seam continuous machine welded.



6.7 Underside of the succeeding sheet and upper side of the PVC membrane already laid should be clean, dry and free from any contamination before welding using automatic welding machine.

6.8 The membrane shall be welded to the pre-installed waterstops to form the compartment.

6.9 Fix **Plasti-board** polypropylene protection boards.

7.0 Testing Welded Seams

Visual Inspection / Non-destructive test:

7.1 Apply force to the upper sheet at the beginning or the end of the test section to tear it away from the lower sheet.

Probe Test (Manual Testing):



7.2 Move the probe or the flat screwdriver along the seam. If any sections are not sealed properly, re-weld them.

Air Pressure Testing (for Double-Welded Seams)



7.3 Check the welded seams with compressed air for seams created by a double-wedge hot-air welder, which produces two parallel welds with a small air channel in between.

7.4 A needle is inserted to pressurize the channel with air (typically to 20-35 psi), and the pressure is monitored for a specified time (e.g., 5 minutes).

7.5 A drop in pressure indicates a leak in one of the two welds, which must then be located and repaired

8.0 Cleaning

8.1 Tools and equipment can be cleaned with water.

Section C: Cautions

Health and safety

- Apply in well-ventilated areas. In closed areas use force ventilation and wear approved respirators when necessary.
- **DonProof L** should not come into contact with skin and eyes. In case of contact with eyes wash immediately with plenty of water and seek medical advice promptly.
- Vapor and atomized liquids are harmful.

Fire:

DonProof L is nonflammable.

For further information on refer to the Material Safety Data Sheet.

Section D: Approval and Variations

This method statement is offered by DCP as a 'standard proposal' for the application of **DonProof L**. It remains the responsibility of the Engineer to determine the correct method for any given application. Where alternative methods are to be used, these must be submitted to DCP for approval, in writing, prior to commencement of any work. DCP will not accept responsibility or liability for variations to the above method statement under any other condition.